CE34000 – Hydraulics
Purdue University – Spring 2026

Course Information

- **Course:** CE34000 - Hydraulics
- **CRN:** 21600
- **Meeting times:** Monday/Wednesdays 1:30-2:20 (ME 1061); Friday either 12:30-1:20 OR 1:30-2:20 (WALC 1132)
- **Instructional Modality** – Face to face
- **Course credit hours –** 3 credits
- **Prerequisites** – C+ or better in CE297/8 (Statics/Dynamics) or equivalent

Course Description

Official course catalog description: Fluid properties; hydrostatics; kinematics and dynamics of fluid flows; conservation of mass, energy, and momentum; flows in pipes and open channels.

Learning Resources, Technology & Texts

- **Required text:**
  - *Munson, Young and Okiishi's Fundamentals of Fluid Mechanics*, 9th Edition (Andrew L. Gerhart, John I. Hochstein, Philip M. Gerhart) - ISBN: 978-1-119-59811-4
- **Software/web resources:** Web browser; Excel, Matlab, or similar
- **Hardware requirements**:
  - PDF scanner (phone is usually fine) for homework problem submission to Gradescope
  - **Hotseat/phone** for class attendance and participation
- **Brightspace learning management system:** The course will be delivered through Brightspace, where you can find all additional course information and materials.
- **Slack:** Homework, quiz, and exam questions will be addressed over Slack, which you can access either as an app on your phone or computer or on the website.

Course Learning Outcomes

1. Apply fundamental concepts and techniques in fluid mechanics in order to perform basic analysis and design of hydraulic systems involving static and flowing fluids.
2. Perform basic fluid mechanics calculations on engineering systems to estimate pressures, velocities, forces, and other properties.
3. Recognize and differentiate between different fluid mechanics concepts in order to analyze hydraulic systems in Civil Engineering.
4. Explain the statics and dynamics of various hydraulic systems found in Civil Engineering in terms of fundamental fluid mechanics concepts.

Assignments / Grading

<u>Exams (65%)</u>		
Exam 1	15%	
Exam 2	15%	
Exam 3	15%	
Final Exam	20%	As scheduled by Purdue during finals week

Other (35%)

Homework	24%	Due weekly, Tuesdays at 11:59pm
Online Quizzes	8%	Due as assigned, typically Fridays
Fri. Attendance	2%	Fridays w/Hotseat quizzes; one absence is allowed
Slack/class particip.	1%	Ask or answer questions on Slack or in class; may also include pop-up attendance polls during M/W lectures.

Grading Scale

Grades are assigned on a +/- scale in this course, using the straightforward breakdown below as a guide.

A+	96.66 – 100	B+	86.66 – 90.00	C+	76.66 – 80.00	D+	66.66 – 70.00
A	93.33 – 96.66	B	83.33 – 86.66	C	73.33 – 76.66	D	63.33 – 66.66
A-	90.00 – 93.33	B-	80.00 – 83.33	C-	70.00 – 73.33	D-	60.00 – 63.33